

DATA VORTEX - Round 1 Phase 1

Social Engine Data Cleaning and Exploratory Data Analysis Report

Source files: Social_Engine_Posts_Corrupted.csv and Social_Engine_Users.csv

1. Objective

Recover the supplied social-media datasets, restore a consistent and analyzable structure, document the cleaning decisions, and derive exploratory insights. The original recovered files are preserved separately from the cleaned outputs.

2. Dataset Overview

Dataset	Rows	Columns
Posts (raw)	12,360	8
Users (raw)	1,500	5
Posts (cleaned)	12,000	8
Users (cleaned)	1,500	5

3. Data Quality Findings and Cleaning Decisions

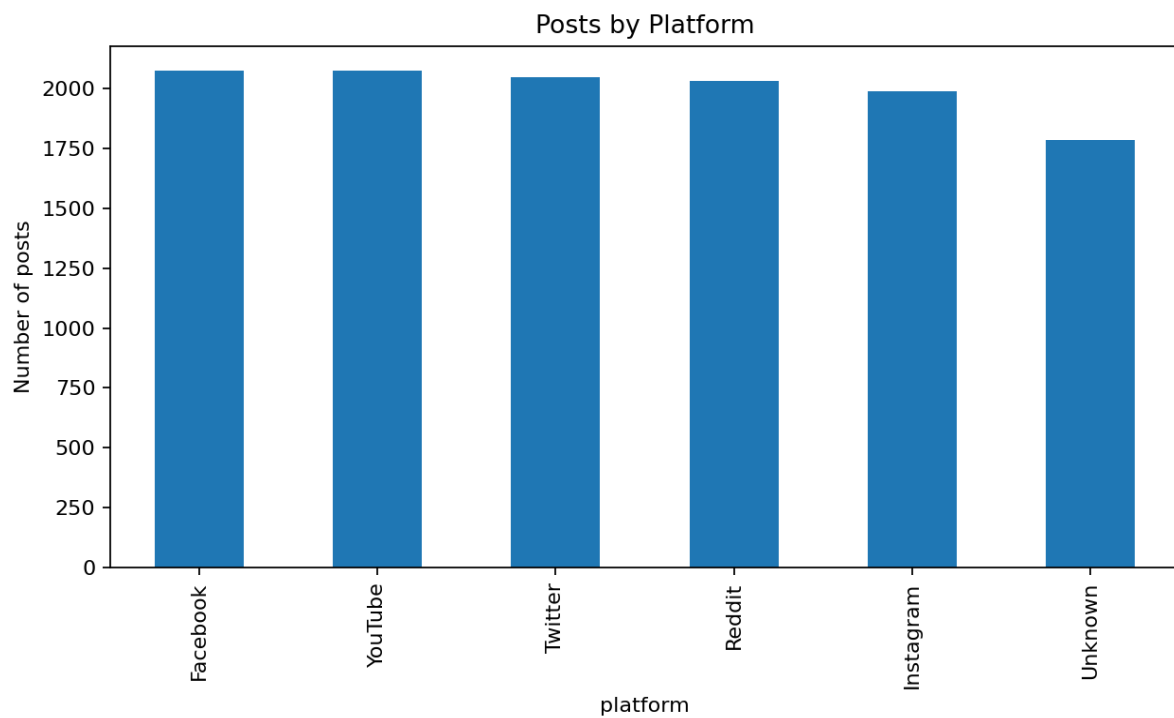
Issue	Finding	Decision
Duplicate rows	360 exact duplicates	Remove duplicate copies; retain first occurrence.
Missing platform	1,846 raw rows	Use Unknown because no reliable platform source exists in Users.
Missing/NULL-style text	1,746 missing + 94 NULL-style placeholders	Treat as missing and set to Unknown.
Missing likes	1,858 raw rows	Median-impute using valid non-negative likes after deduplication.
Negative likes	525 raw rows	Treat as invalid/missing, then median-impute.
Timestamp formats	3 formats across all 12,360 raw rows	Standardize to YYYY-MM-DD HH:MM:SS.
User references	0 missing user IDs	No referential repair required.
Shares/comments	0 negative values	No correction required.

4. Cleaning Validation

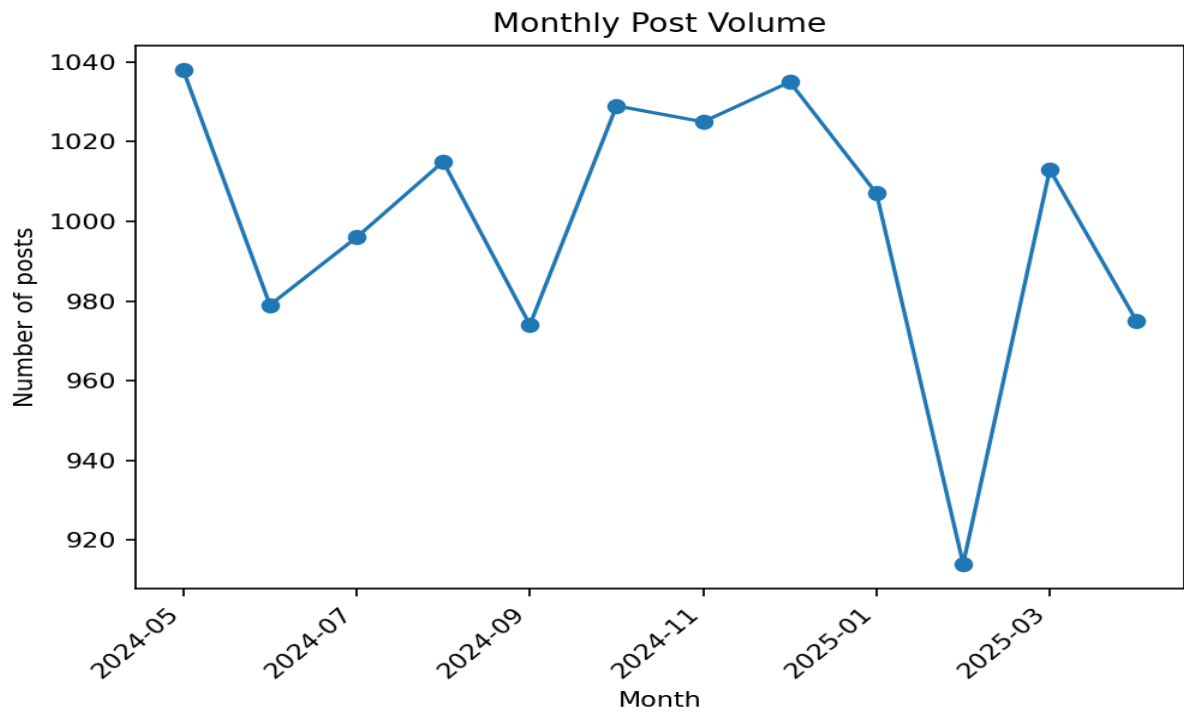
- Cleaned posts: 12,000 rows
- Exact duplicates remaining: 0
- Unique post IDs: 12,000
- Unknown platform records: 1,784
- Unknown/invalid text records: 1,779
- Missing likes after cleaning: 0
- Negative likes after cleaning: 0
- Unparsed timestamps after cleaning: 0
- Post user IDs missing from Users: 0

5. Exploratory Data Analysis

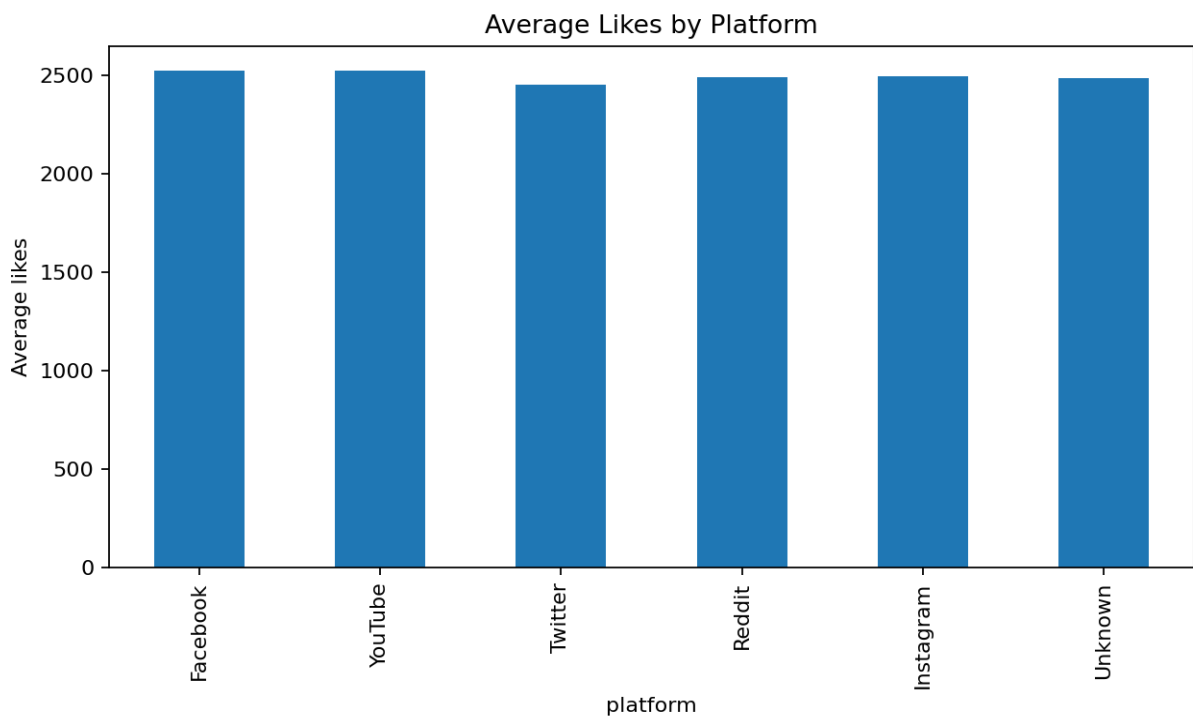
The cleaned dataset contains 12,000 unique posts from 1,500 users. The observed date range is 2024-05-01 to 2025-04-30 21:57:10. Average likes are 2,495.1, average shares are 1,007.2, and average comments are 504.3.



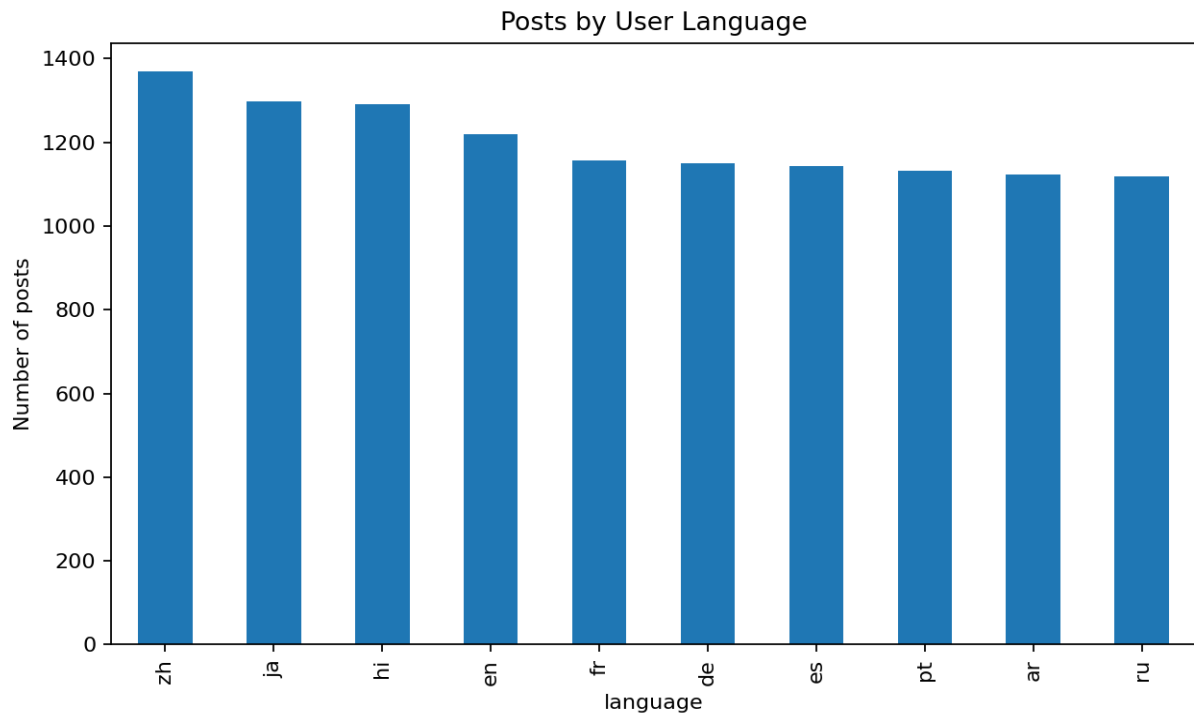
Facebook and YouTube have the highest post counts among identified platforms. 1,784 cleaned records remain in the Unknown platform category, which transparently reflects unrecoverable source information.



May 2024 has the highest monthly post volume at 1,038 posts. Volume remains substantial across the period.



Average likes are relatively close across the five named platforms; therefore platform volume and average likes should be interpreted separately.



Chinese (zh) is the most represented user language, followed by Japanese (ja) and Hindi (hi). The dataset therefore supports multilingual behavioral segmentation in later analysis.

6. Key Insights

1. Recovery produced two linked datasets: 12,360 raw post rows and 1,500 user rows.
2. Removing 360 exact duplicates produced 12,000 unique posts.
3. Missing platform and text values are retained transparently as Unknown instead of being guessed.
4. Negative and missing likes were handled using a valid-like median of 2,502.
5. All timestamps were successfully classified into three source formats and standardized.
6. All post user IDs matched the Users table, so the post-user relationship is structurally consistent.

7. Reproducibility Note

The bundle includes raw recovered files, cleaned CSVs, the cleaning log, EDA outputs, plots, and a Jupyter notebook/script. The cleaning rules are deterministic and can be rerun from the raw files.